

Ahhyun Lucy Lee

📍 Seoul, Korea ✉ ahhyun724@snu.ac.kr ☎ +821049099247 in Ahhyun Lucy Lee 🌐 yahyune

Education

PhD	Seoul National University , Brain and Cognitive Sciences	Sep '25 – Current
BS	POSTECH , Life Science Major, Computer Science and Engineering Minor	Mar '19 – Aug '24
	UIUC , Department of Psychology, Champaign, IL, USA <i>Non-degree exchange semester</i>	Jan '23 – May '23

Conferences

DIVER-0 : A Fully Channel Equivariant EEG Foundation Model Danny Dongyeop Han, Ahhyun Lucy Lee, Taeyang Lee, Yonghyeon Gwon, Sebin Lee, Seongjin Lee, David Keetae Park, Shinjae Yoo, Jiook Cha, Chun Kee Chung arXiv:2507.14141	<i>ICML 2025 Workshop on GenBio (Spotlight)</i>
Enhancing psychiatric disorder diagnosis: integrating multilabel metadata with contrastive learning on T1 structural MRI Danny Dongyeop Han, Ahhyun Lucy Lee, Junbeom Kwon, Jiook Cha	<i>OHBM2025</i>
The Impact of Low Motion Smoothness on Auditory Processing Ahhyun Lucy Lee, Jinhee Kim, You Jeong Hong, Kyung Myun Lee, Kyogu Lee	<i>ICMPC17</i>

Publications

Exploring the diagnostic potential of EEG theta power and interhemispheric correlation of temporal lobe activities in Alzheimer's Disease through random forest analysis. Ahhyun Lucy Lee, Eunjin Hwang & Jeongeun Hwang <i>Computers in biology and medicine</i> , 192, 110248. DOI: 10.1016/j.compbiomed.2025.110248	2025
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Skills

Languages: English (Professional, TOEFL ibt 107/120), Korean (Native)

Programming Language: Python, R, Matlab, C, C++

Relevant Coursework Highlights:

- **Neuroscience:** Language and the Brain, Developmental Cognitive Neuroscience, Principles of Neuroscience
- **Computer Science:** Machine Learning and Deep Learning, Machine learning for Computer Vision, Datascience for Natural Language Processing

Research & Internship Experience

Internship in Lablup (Opensource cloud computing and AI service company) • Conducted research on the topic "Exploring the Diagnostic Potential of EEG Theta Power and Interhemispheric Correlation of Temporal Lobe Activities in Alzheimer's Disease through Random Forest Analysis"	Seoul, Korea Jan '24 – Present
Internship in MARG (Music and Audio Research Group, SNU) • Conducted independent research on "Effect of Beat Coordination of Simultaneous Dance Stimuli on Sound Perception in Audiovisual Dance," developing a real-world observation into a novel research question.	Suwon, Korea Jul '21 – Aug '23

- Designed and implemented experimental paradigm using PsychToolbox in MATLAB and performed statistical analysis in R.
- Presented research findings at the 17th International Conference on Music Perception and Cognition (ICMPC17).

Internship in Selexcine (Biopharmaceutical development company specialized in field of immunotherapeutics)

Pohang, Korea
Jul '20 – Aug '20 &
Jan '21 – Feb '21

- Studied Immunology and practiced experimental techniques. Conducted experiments to observe the correlation between tumor size and immune cell composition difference according to TCB2 injection.
- Experienced the initial step of developing new medicine; searching for possible candidate. Wrote summarized report of 40 research papers related to CD73.

Antibody Structure and Engineering, POSTECH, Research Participation

Pohang, Korea
Feb '21 – Jun '21

- Practiced protein structure analysis using simulation tools such as Pymol, maestro, discovery studio.

Molecular Neuropsychiatry Lab, POSTECH, Freshman Research Participation

Pohang, Korea
Jun '19 – Aug '19

- Observed insertion of plasmid DNA into the embryonic mouse brain and separating the brain from in utero electroporation, observed mouse brain using a microscope, exercised dehydration and cryosection.

Neuroepigenetics Lab, POSTECH, Freshman Research Participation

Pohang, Korea
Mar '19 – Jun '19

- Observed and learned mouse hippocampal primary cell culture and observation through the microscope.

Honors and Awards

National Science and Engineering Undergraduate Scholarship

Mar '19 – Aug '24

- Awarded full-ride national scholarship (KOSAF) for future scientists and engineers based on high school academic excellence.
- Maintained scholarship throughout undergraduate program with consistent strong academic performance.

Genexine Research Award

Apr '21

- Awarded to top undergraduate researchers for outstanding achievement.

Highlighted Projects

Representation of previous research regarding effect of imagination on auditory loudness perception (SOSC441) Final Team Project

Pohang, Korea
Jun '20 – Jul '20

- Presented the research idea to team members based on previous research. Organized the whole process, conducting experiment and analyzing results. Found reverse effect compared to original study and analyzed the reason.

CapStone Project in Google Data Analytics Certificate Program

Aug '23

Leadership & Extracurricular Activities

BCSC (Brain Cognitive Science Community), President

Korea
Sep '21 – Feb '23

- Planned and held Semestral conference.
- Learned to devise the scheme of academic event and communicate with club members.

POSTECH, Department of Life Science, 33rd Treasurer

Pohang, Korea
Mar '21 – Dec '21

- Drew up a budget and bill statement.

SMP(Student Mentoring Program), General Life Science (LIFE101), Student Mentor

- Taught General Life Science to freshmen.

Pohang, Korea
Sep '20 – Dec '20 &
Feb '21 – Jun '21 &
Feb '22 – Jun '22

Ctrl-D (Street Dance Club), Member

- Participated in various dance performances across campus and community events.
- Actively developed strategies to maintain club engagement and activities during COVID-19 restrictions.
- Served on Sexual Harassment and Sexual Violence Response Task Force, monitoring and ensuring compliance with organizational bylaws.

Pohang, Korea
Mar '19 –

DAIS (Data Science and Machine Learning Club), Member

- Developed a personalized vitamin recommendation system based on dietary habits, height, and weight as part of a collaborative team project.
- Conducted literature review on nutritional science and built data analysis tools using R.

Champaign, IL, USA
Jan '23 – May '23

Imagination Dance Crew, Member

- Performed in 2 performances and one competition throughout a semester. Choreographed on one K-pop song.

Champaign, IL, USA
Jan '23 – May '23